

Phase 2a – Live Prices & Algeria Project Map Interactive

Version: 1.0 **Date:** 2026-06-21 **Status:** Planning — Ready for Development **PRDs:**
01_LIVE_ENERGY_PRICE.md · 05_ENERGY_PROJECTS_ALGERIA.md **Index:** ← [Back to PRD Index](#)

Table of Contents

- 1. [Tech Stack](#)
- 2. [Diagram Flow](#)
- 3. [Design Layout](#)
- 4. [Timeline](#)

1. Tech Stack

1.1 Shared Foundation (Already Built)

LAYER	TECHNOLOGY	VERSION	NOTES
Framework	Next.js (App Router)	16.2.6	SSR + API routes
Language	TypeScript	^5	Strict mode
Styling	Tailwind CSS	^4	Custom theme (navy, teal, slate)
Database	Supabase PostgreSQL	—	<code>pgvector</code> extension available
Auth	Supabase Auth	^2.106	SSR helpers via <code>@supabase/ssr</code>
Icons	Lucide React	^1.16	
Vercel	Deployment + Cron	—	Hosting + scheduled jobs

1.2 Feature 1 — Live Energy Prices (F01)

LAYER	TECHNOLOGY	PURPOSE
API Route	Next.js Route Handler	GET /api/energy-prices — provider-agnostic facade
Providers	Multi-source abstraction	PriceProvider interface → pluggable sources
Primary Source	Argus Media / S&P Global Platts API	Saharan Blend, Brent, WTI, Murban, TTF (subscription required)
Validation	OPEC (opec.org)	OPEC Basket cross-check — public/free
Fallback	OilPrice.com scraping	Secondary reference — cheerio
Supplemental	EIA API (eia.gov)	Henry Hub (free API)
Cache	In-memory Map with TTL	5 min tiered cache (server-side)
Client Polling	SWR	^2.2.0 — 5-min refreshInterval
Historical DB	energy_prices table (Supabase)	Every fetch persisted — 90d per-fetch, 2yr hourly, indefinite daily
Benchmark Registry	Config-driven BENCHMARK_REGISTRY	Add benchmark → auto-flows to API + DB + UI + intelligence pages
Intelligence Pages	/intelligence/[benchmark]	Per-benchmark pages with price, trends, AI analysis, sources
Charts	Recharts or Chart.js (lightweight)	Historical trend charts (7d/30d/90d/1y)
Homepage Widget	PriceTicker component	Horizontal ticker — top 4 benchmarks
Components	PriceGrid + PriceCard	Full price dashboard on /energy-prices

New packages to install:

```
{  
  "swr": "^2.2.0",  
  "cheerio": "^1.0.0",  
  "recharts": "^2.12.0"  
}
```

1.3 Feature 2 — Algeria Project Map Interactive (F05)

LAYER	TECHNOLOGY	PURPOSE
Crawlers	Vercel Cron + cheerio + @mozilla/readability	Scheduled crawl of 19 sources every 6-12h
AI Extraction	OpenAI (gpt-4o-mini)	LLM extracts 18-field structured JSON from unstructured pages (FR/AR/EN)
Deduplication	Embedding similarity (cosine ≥ 0.85) + GPS proximity (5 km) + fuzzy name matching	Match incoming projects against existing DB
Change Detection	Field-level diff comparison	Detect NEW / UPDATED / UNCHANGED / CONFLICT
Review CMS	Admin route /admin/projects/review	Approve / edit / reject / resolve conflicts
Database	energy_projects table (Supabase)	18-field schema + audit trail (sync_history JSONB)
Map	Leaflet + React-Leaflet	Interactive map with GPS markers, popups, clustering
Frontend	/energy-projects page	Filter bar (category pills + status dropdown + wilaya + search) + project grid + map
Project Cards	ProjectCard component	Status badge, category, capacity, operators, source attribution
Detail View	Modal or dedicated page	All 18 fields, source URL, last-updated timestamp
i18n	Existing i18n system (src/lib/i18n/)	FR / EN / AR — project content in mixed languages

New packages to install:

```
{
  "leaflet": "^1.9.0",
  "react-leaflet": "^4.2.0",
  "@types/leaflet": "^1.9.0",
  "cheerio": "^1.0.0",
  "@mozilla/readability": "^0.5.0",
  "gpt-tokenizer": "^2.1.0"
}
```

1.4 Shared Infrastructure

COMPONENT	TECHNOLOGY	USED BY
Crawl scheduling	Vercel Cron	F05 crawlers
LLM calls	OpenAI (gpt-4o-mini)	F05 AI extraction
Embeddings	OpenAI text-embedding-3-small	F05 deduplication
Token tracking	EXTRACTION_DAILY_MAX_TOKENS=300000	F05 budget cap
Content fetch	cheerio + readability	F05 page scraping

1.5 Environment Variables (New)

```
# F01 – Live Prices
# EIA_API_KEY=*** (future: EIA API v2)

# F05 – Project Extraction
EXTRACTION_DAILY_MAX_TOKENS=300000
OPENAI_EMBEDDING_MODEL=text-embedding-3-small
```

1.6 Frontend Component Tree

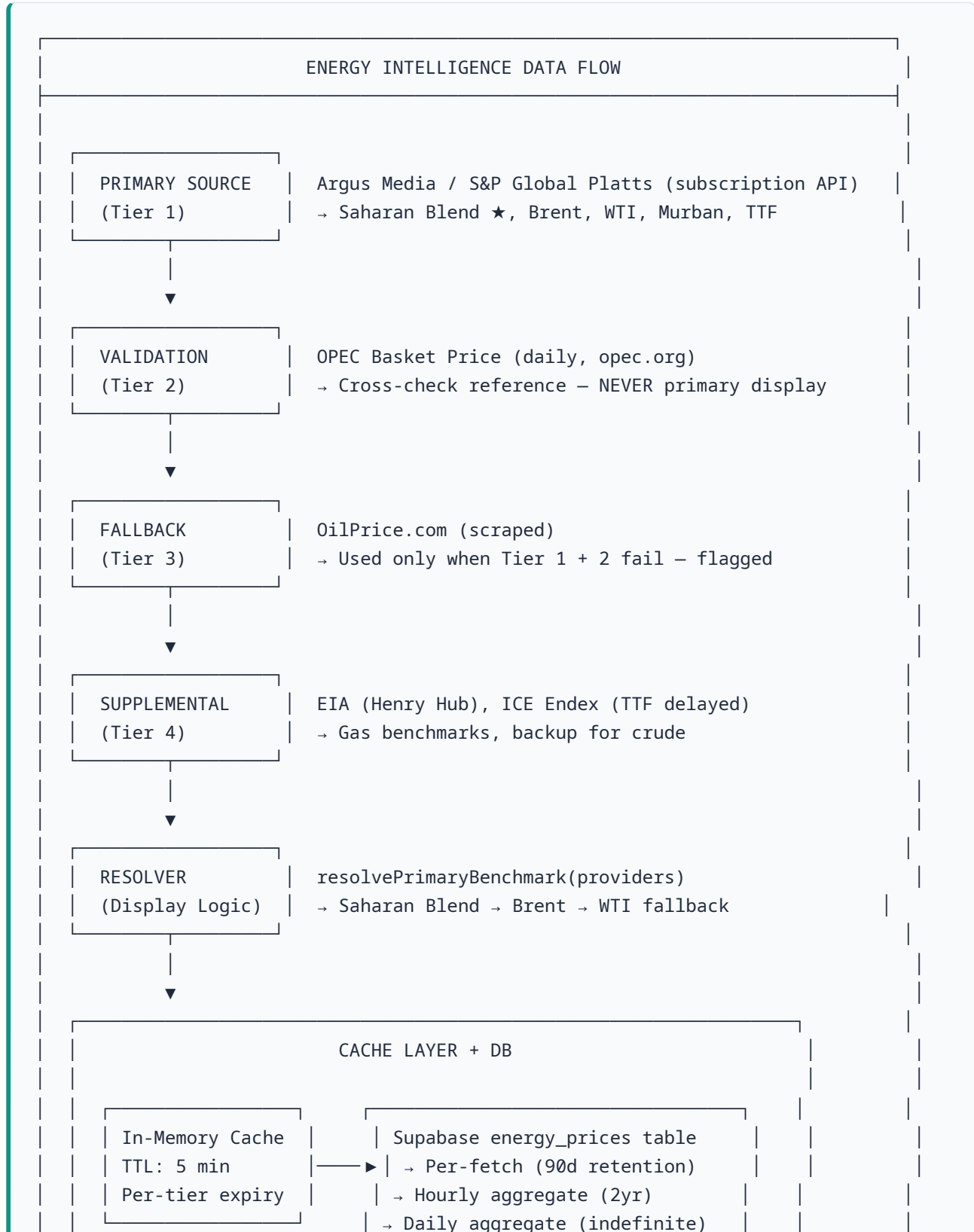
```

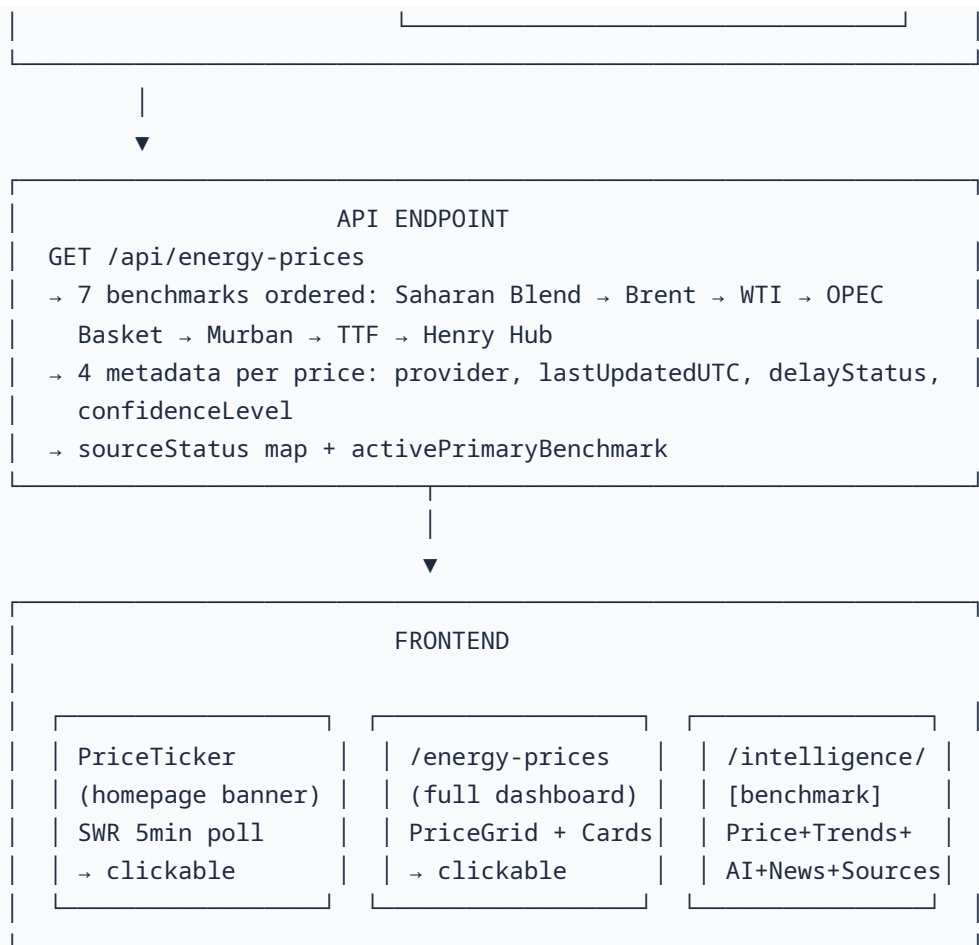
src/
├── app/
│   ├── (site)/
│   │   ├── energy-prices/
│   │   │   └── page.tsx          ← NEW: Price dashboard
│   │   ├── energy-projects/
│   │   │   ├── page.tsx        ← NEW: Project listing + map
│   │   │   └── [slug]/
│   │   │       └── page.tsx    ← NEW: Project detail
│   │   └── intelligence/
│   │       └── [benchmark]/
│   │           └── page.tsx    ← NEW: Intelligence page per benchmark
│   ├── admin/
│   │   └── (auth)/
│   │       ├── projects/
│   │       │   ├── review/
│   │       │   │   └── page.tsx ← NEW: Review queue
│   │       │   └── page.tsx    ← NEW: Project list management
│   └── api/
│       ├── energy-prices/
│       │   └── route.ts        ← NEW: Prices API endpoint
│       ├── energy-projects/
│       │   ├── route.ts      ← NEW: Projects API
│       │   ├── crawl/
│       │   │   └── route.ts   ← NEW: Cron-triggered crawl
│       │   └── review/
│       │       └── route.ts   ← NEW: Review actions API
│       └── intelligence/
│           └── [benchmark]/
│               └── route.ts   ← NEW: Intelligence data API
├── components/
│   ├── prices/                ← NEW: Price components
│   │   ├── PriceTicker.tsx
│   │   ├── PriceGrid.tsx
│   │   ├── PriceCard.tsx
│   │   ├── PriceChart.tsx
│   │   ├── BenchmarkIntelligence.tsx
│   │   ├── SourceAttribution.tsx
│   │   └── PriceMetadata.tsx
│   └── projects/              ← NEW: Project components
│       ├── ProjectGrid.tsx
│       ├── ProjectCard.tsx
│       ├── ProjectMap.tsx
│       ├── ProjectDetail.tsx
│       └── ProjectFilters.tsx
  
```

```
| | | ProjectStatusBadge.tsx
| | | ReviewQueue.tsx
| | └─ admin/
| | | ProjectReviewCard.tsx ← NEW
| | | ProjectReviewForm.tsx ← NEW
| └─ lib/
| | prices/ ← NEW: Prices logic
| | | benchmark-registry.ts
| | | providers/
| | | | types.ts
| | | | argus-platts.ts
| | | | opec.ts
| | | | eia.ts
| | | └─ oilprice.ts
| | | cache.ts
| | | resolver.ts
| | | historical.ts
| └─ projects/ ← NEW: Projects logic
| | crawlers/
| | | tier1-algerian.ts
| | | tier2-international.ts
| | | tier3-news.ts
| | | └─ tier4-companies.ts
| | | extractor.ts
| | | deduplicator.ts
| | | schema.ts
| | └─ sync-history.ts
```

2. Diagram Flow

2.1 Live Energy Prices — Architecture & Data Flow





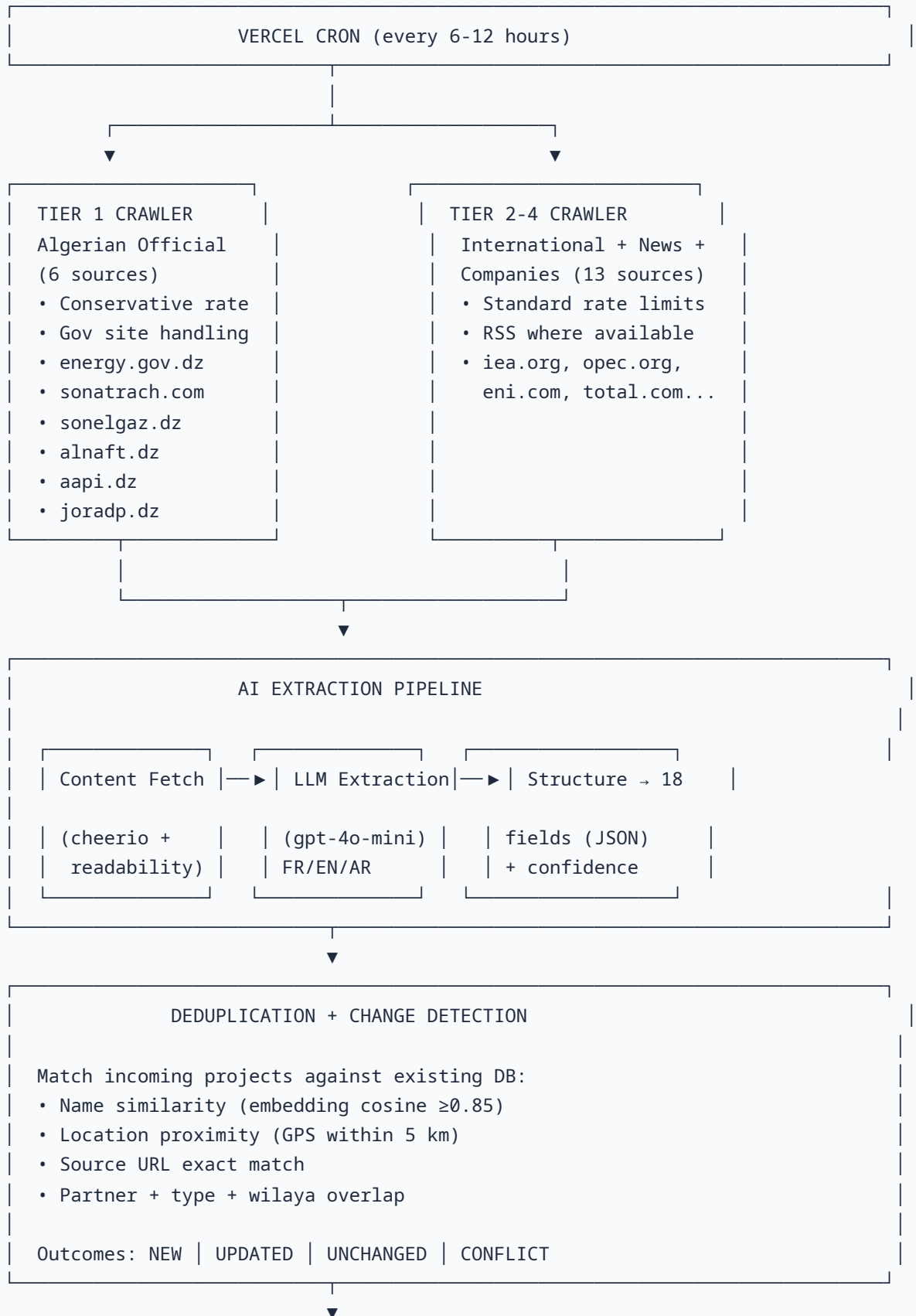
FALLBACK DISPLAY LOGIC:

Saharan Blend available → ★ Saharan Blend is primary display
Saharan Blend down → Brent becomes primary display (NOT OPEC Basket)
OPEC Basket → Always validation/reference – never promoted
⚠ OilPrice.com is NEVER the primary source for pricing

2.2 Live Energy Prices — Request Flow



2.3 Algeria Projects — Automated Pipeline



REVIEW QUEUE (Admin CMS)

NEW PROJECTS (approve/reject)	CHANGES TO REVIEW (approve/dismiss)	CONFLICTS (resolve)
----------------------------------	--	------------------------

Admin actions: Approve → Published | Edit → Published
Reject → Archived | Dismiss change → Keep prev



LIVE DATABASE (Supabase – energy_projects)

18-field schema + source_url + source_urls[] + sync_history JSONB
Audit trail: every create/update/dismiss tracked
Stale flagging: >30 days without source update



FRONTEND

/energy-projects

- Hero: "Algeria Energy Projects Database – auto-updated from 19 sources"
- Filter bar: category pills + status dropdown + wilaya + search
- Project grid: cards with status/category badges, capacity, operators
- Interactive map: Leaflet + React-Leaflet with GPS markers + popups
- Project detail: modal or dedicated page with all 18 fields

/admin/projects/review

- Tabs: New | Changes | Conflicts
- Review cards: project summary, source, confidence
- Actions: Approve / Edit / Reject / Dismiss / Resolve

2.4 Database Schema — New Tables

energy_prices	
id	BIGSERIAL PRIMARY KEY
benchmark	VARCHAR(64) NOT NULL
category	VARCHAR(32) NOT NULL (crude, natural_gas, ...)
value	DECIMAL(12,4) NOT NULL
unit	VARCHAR(16) NOT NULL (USD/bbl, EUR/MWh, ...)
change	DECIMAL(8,4)
change_pct	DECIMAL(6,2)
trend	VARCHAR(4) (up, down, flat)
provider	VARCHAR(32) NOT NULL
provider_tier	SMALLINT NOT NULL (1, 2, 3, 4)
delay_status	VARCHAR(20) NOT NULL
confidence	VARCHAR(8) NOT NULL (high, medium, low)
availability	VARCHAR(24) NOT NULL DEFAULT 'available'
source_url	TEXT
fetches_at	TIMESTAMPTZ NOT NULL DEFAULT NOW()
is_active_primary	BOOLEAN DEFAULT FALSE
INDEXES: idx_ep_benchmark_time, idx_ep_category_time, idx_ep_fetches_at	
MATERIALIZED VIEW: energy_prices_latest (last 2h, per benchmark)	

energy_projects	
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()
name	TEXT NOT NULL
slug	TEXT NOT NULL UNIQUE
energy_sector	TEXT NOT NULL (oil, gas, lng, renewable...)
project_type	TEXT (pipeline, field, plant, ...)
wilaya	TEXT
municipality	TEXT
gps_lat	DECIMAL(10,7)
gps_lng	DECIMAL(10,7)
operating_company	TEXT
project_partners	TEXT[] DEFAULT '{}'
investment_value	TEXT
production_capacity	TEXT
project_status	TEXT NOT NULL (operational, construction...)
completion_pct	DECIMAL(5,2)
start_date	DATE
expected_completion	DATE

project_description	TEXT
source_url	TEXT
source_name	TEXT
last_updated	TIMESTAMPTZ DEFAULT NOW()
last_source_check	TIMESTAMPTZ DEFAULT NOW()
source_urls	TEXT[] DEFAULT '{}'
review_status	TEXT NOT NULL DEFAULT 'pending'
reviewed_by	TEXT
reviewed_at	TIMESTAMPTZ
featured	BOOLEAN DEFAULT false
sort_order	INTEGER DEFAULT 0
sync_history	JSONB DEFAULT '[]'
created_at	TIMESTAMPTZ DEFAULT NOW()
updated_at	TIMESTAMPTZ DEFAULT NOW()
INDEXES: idx_eproj_sector, idx_eproj_status, idx_eproj_wilaya, idx_eproj_review, idx_eproj_source	

3. Design Layout

3.1 Design System — Existing Theme

TOKEN	VALUE	USAGE
--navy	#0f172a	Primary text, headers
--green	#0d9488	Accent, buttons, CTAs
--line	#e2e8f0	Borders, dividers
--muted	#64748b	Secondary text, metadata
--bg	#f8fafc	Page background
Font (heading)	Serif (Newsreader/Inter)	Headings, hero text
Font (body)	Sans (system)	Body text, cards, UI

3.2 Feature 1 — Live Energy Prices UI

3.2.1 Homepage Price Ticker

Live Energy Markets			
★ Saharan Blend	Brent	WTI	OPEC Basket
\$94.20 ▼0.9%	\$93.68 ▼1.2%	\$89.45 ▲0.3%	\$92.10 ▼0.8%
Platts · RT · High	Platts·RT	EIA·15min	OPEC·Daily
[Updated 2 min ago]		[View All Prices →]	

Design specs:

- Horizontal scrolling ticker (smooth CSS scroll or manual)
- 4-5 benchmarks visible at once on desktop
- Each benchmark: name, price, change %, trend arrow, provider badge
- Green background section with white/light text
- Click any benchmark → /intelligence/[benchmark]
- Mobile: 2 visible, horizontal swipe
- Auto-scroll every 8 seconds, pause on hover
- Last updated timestamp bottom-left

3.2.2 /energy-prices — Full Price Dashboard

Breadcrumb: Home > Energy Prices

Energy Intelligence Platform
Real-time energy commodity prices from authoritative sources.
Updated every 5 minutes. Last update: 2026-06-21 14:30 UTC

★ PRIMARY BENCHMARK

Saharan Blend

Saharan Blend

\$94.20

★ Algerian Crude

▼ 0.89%
-\$0.85

Source: Platts · Real-time · High confidence
Last updated: 2026-06-21 14:30 UTC

[View Intelligence →]



Brent

\$93.68
▼1.2% -\$1.14

Platts · RT · High
[Intelligence →]



WTI

\$89.45
▲0.3% +\$0.27

EIA · 15min · High
[Intelligence →]



OPEC Basket

\$92.10
▼0.8% -\$0.74

OPEC · Daily
· Med conf



Murban

\$91.55
▼0.5% -\$0.46

Platts · RT · High
[Intelligence →]



TTF (Natural Gas)

€28.40/MWh
▲1.8% +€0.50

ICE · 30min · Med
[Intelligence →]



Henry Hub

\$3.15/MMBtu
▼0.7% -\$0.02

EIA · Daily
· Med conf

Source Status: Platts  | Argus  | OPEC  | EIA  | OilPrice 

 Saharan Blend temporarily unavailable – Brent shown as primary.
OPEC Basket shown for validation.

Design specs:

- Primary benchmark card: larger, highlighted border (teal/green accent)
- Standard cards: 2x3 grid on desktop, 1-column on mobile
- Each card: benchmark icon, name, price (large), change (colored), metadata
- Change: green ▲ for up, red ▼ for down
- Each card clickable → /intelligence/[benchmark]
- Source status bar at bottom
- Responsive: 3-col desktop → 2-col tablet → 1-col mobile

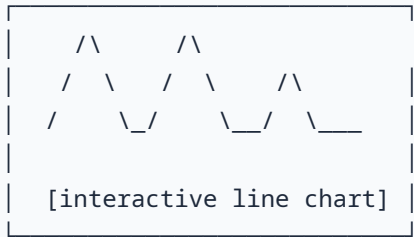
3.2.3 /intelligence/[benchmark] — Intelligence Page

Breadcrumb: Home > Energy Prices > Saharan Blend Intelligence

★ SAHARAN BLEND \$94.20 ▼0.9% · Platts · Real-time · High confidence
Algerian crude benchmark – last updated 14:30 UTC

HISTORICAL TREND

7d · 30d · 90d · 1y [tabs]



AI MARKET ANALYSIS

"Saharan Blend declined 0.9% today, driven by..."

Sources: OPEC MOMR (Jun 2026),
Platts Crude Oil Marketwire,
Sonatrach PR (Jun 15)

[Expand Analysis →]

RELATED NEWS & ANALYSIS

[View All →]

- Sonatrach announces new production targets for Hassi Messaoud...
sonatrach.com · 2 days ago
- OPEC+ maintains output cuts through Q3 2026...
reuters.com · 3 days ago
- Algerian crude exports to Europe rise 2.3% in May...
energy.gov.dz · 5 days ago

PRODUCTION & COUNTRY CONTEXT

Algeria crude production: ~960,000 bpd (OPEC MOMR Jun 2026)
Main export destinations: Italy, France, Spain, Netherlands
Key infrastructure: MEDGAZ Pipeline, TransMed Pipeline
Saharan Blend API gravity: ~45° | Sulfur: ~0.10%

[View Algeria Energy Projects →]

RELATED BENCHMARKS

Brent \$93.68 ▼1.2% | OPEC Basket \$92.10 ▼0.8% | Murban \$91.55

SOURCES & REFERENCES

AI analysis used:

- OPEC Monthly Oil Market Report (June 2026)
- Platts Crude Oil Marketwire (June 21, 2026)
- Sonatrach press release (June 15, 2026)

Official reports & publications:

- IEA Oil Market Report – June 2026
- EIA Short-Term Energy Outlook – June 2026
- Algeria Ministry of Energy – Production Statistics 2025

3.3 Feature 2 — Algeria Project Map Interactive UI

3.3.1 /energy-projects — Main Page

Breadcrumb: Home > Algeria Energy Projects

Algeria Energy Projects Database

Comprehensive, self-updating database of energy infrastructure projects across Algeria. Sourced from 19 official channels. Last sync: 2 hours ago.
[How This Works ↓]

Filter:

[All] [Oil & Gas] [Renewable] [LNG] [Power] [Hydrogen]

Status: [All ▼] Wilaya: [All ▼]

● Operational

📦 Pipeline

Medgaz Gas Pipeline
Algeria → Spain (Almería)
10 bcm/year capacity
Sonatrach · Naturgy

🔗 Source: sonatrach.com

🕒 Updated: 2026-06-19

[View Details →]

● Under Construction

☀️ Solar

Solar 15 GW Programme
National renewable rollout
15 GW target capacity
Sonelgaz · Sonatrach

🔗 Source: energy.gov.dz

🕒 Updated: 2026-06-16

[View Details →]

● Planned

📦 Pipeline

Trans-Saharan Gas Pipeline
Nigeria → Algeria → Europe
30 bcm/year planned
Sonatrach · NNPC · Niger

🔗 Source: energy.gov.dz

● Operational

🔥 LNG

Skikda LNG Terminal
Mediterranean export hub
Multiple trains
Sonatrach

🔗 Source: sonatrach.com

Page 19 of 40
Page 19 of 40

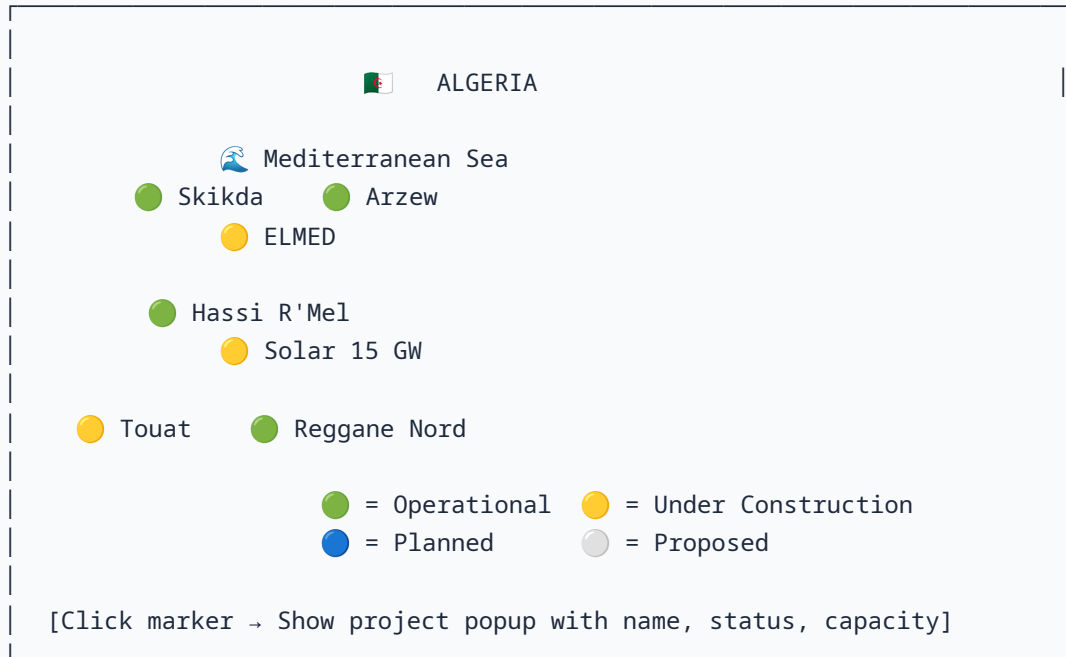
Updated: 2026-06-10

[View Details →]

Updated: 2026-06-15

[View Details →]

INTERACTIVE PROJECT MAP



Design specs:

- Filter bar: category pills (rounded, active state with green fill), dropdowns, search input
- Project cards: 3-column grid on desktop, 2 on tablet, 1 on mobile
- Each card: status badge (colored), category icon, project name, route/capacity, operators, source
- Map: full-width, ~400px height, Leaflet with OpenStreetMap tiles
- Map markers: colored by status (green=operational, amber=construction, blue=planned, gray=proposed)
- Map popups: project name, status badge, capacity, click to view details
- Project detail: expand inline or modal overlay with all 18 fields
- Responsive: map collapses below filter on mobile

3.3.2 Project Detail Card (Expanded/Modal)

Medgaz Gas Pipeline

[× Close]

● Operational 📦 Pipeline

Overview

Direct subsea natural gas pipeline connecting the Hassi R'Mel field in Algeria to Almería, Spain. Operational since 2011, recently expanded from 8 bcm/year to 10 bcm/year capacity. Strategic infrastructure for Algerian gas exports to Southern Europe.

Location

Beni Saf, Aïn Témouch
→ Almería, Spain
Wilaya: Aïn Témouchent

Capacity

10 bcm/year
(expanded from 8)

Operator

Sonatrach

Partners

- Naturgy (Spain)
- Sonatrach (Algeria)

Start Date

2008

Investment

~€900 million

 Primary Source: sonatrach.com/press/medgaz-expansion-2026

 Additional Sources: eni.com, reuters.com

 Last updated: 2026-06-19 08:30 UTC (source check: 2026-06-20)

Sync History:

- 2026-06-19 – Capacity updated: 8 → 10 bcm/year (source: sonatrach.com)
- 2026-06-15 – Status confirmed: Operational (source: eni.com)

- 2026-06-10 – Project created (source: sonatrach.com)

3.3.3 Admin Review Queue — /admin/projects/review

Projects – Review Queue

12 pending

[ New (5)] [ Changes (4)] [ Conflicts (3)]

 NEW PROJECT

Sonatrach · 2h ago

Hassi R'Mel Solar Plant

Sector: Renewable | Type: Solar | Wilaya: Laghouat

Capacity: 150 MW | Status: Under Construction

Operator: Sonelgaz | Partners: Sonatrach, Eni

Description: New solar PV plant at Hassi R'Mel site, part of national renewable energy programme. Expected completion 2028.

 Source: sonatrach.com/press/hassi-rmel-solar

Confidence: High (8/18 fields high confidence)

[Approve] [Edit] [Reject]

 CHANGES DETECTED

Eni · 5h ago

Medgaz Pipeline Expansion

- Capacity: 8 bcm/year → 10 bcm/year
- Status: Operational → Operational (unchanged)

 Source: eni.com/media/press-release/medgaz-expansion

[Approve Change] [Dismiss] [Edit]

 CONFLICT

Reuters · 1d ago

TSGP Pipeline – Status Discrepancy

Sonatrach (Tier 1): Planned

Reuters (Tier 3): Under Discussion

JORADP (Tier 1): Proposed

Eni (Tier 4): Planned

[Keep Sonatrach (Planned)] [Accept Reuters (Under Discussion)]

[Manual Resolve →]

3.4 Status Badge System

STATUS	COLOR	BADGE	CSS CLASS
Operational	Green	 Operational	bg-green-100 text-green-800
Under Construction	Amber	 Under Construction	bg-amber-100 text-amber-800
Planned	Blue	 Planned	bg-blue-100 text-blue-800
Proposed	Gray	 Proposed	bg-gray-100 text-gray-600
Completed	Dark Green	 Completed	bg-emerald-100 text-emerald-800
Suspended	Red	 Suspended	bg-red-100 text-red-800

3.5 Public-Facing Messaging Rules

INTERNAL STATUS	USER-FACING DISPLAY
Premium API not subscribed	"Temporarily unavailable"
API timeout / fetch failed	"Updating — check back shortly"
All sources down	"Prices temporarily unavailable — last updated [timestamp]"
Benchmark not yet configured	"Coming soon"

 Never expose internal licensing status or infrastructure details to end users.

3.6 Responsive Breakpoints

BREAKPOINT	LAYOUT
Desktop (≥1024px)	3-col price grid, 3-col project grid, side-by-side intelligence layout, full map width
Tablet (≥640px)	2-col price grid, 2-col project grid, stacked intelligence layout, full map width
Mobile (<640px)	1-col everything, map height 300px, filter bar scrollable, ticker shows 2 items, cards full-width

3.7 Loading & Empty States

Loading states:

- Price cards: skeleton shimmer (pulse animation)
- Project cards: skeleton shimmer
- Charts: skeleton area with "Loading historical data..."
- Map: skeleton rectangle with "Loading map..."

Empty states:

- No projects match filters: "No projects match your criteria. Try adjusting filters."
- All prices unavailable: warning banner + last known values with stale timestamp
- Review queue empty: "All caught up! No projects pending review."

4. Timeline

4.1 Overview

PHASE	FEATURE	DAYS	DEPENDENCIES
Setup	Foundation + DB Schema	2	None
F01	Live Energy Prices	5	Setup
F05	Algeria Project Map	18-22	Setup
Total	With overlap	~20-24 days	

F01 and F05 can overlap significantly — they share no dependencies except the foundation setup.

4.2 Week-by-Week Breakdown

WEEK 1 – Foundation + F01 Start

```
| Day 1-2: Foundation |
| └─ Database migrations (energy_prices, energy_projects tables) |
| └─ Base types (BenchmarkPrice, ProjectSchema, PriceProvider) |
| └─ Install packages: swr, cheerio, recharts, leaflet, etc. |
| └─ Benchmark registry config |
|
| Day 3-5: F01 – API + Data Layer |
| └─ PriceProvider interface + types |
| └─ Provider implementations (OPEC, EIA, OilPrice fallback) |
| └─ Resolver logic (Saharan Blend → Brent fallback) |
| └─ In-memory cache with 5-min TTL |
| └─ GET /api/energy-prices endpoint |
| └─ Historical DB write on every fetch |
```

WEEK 2 – F01 Frontend + F05 Crawler

```
| Day 6-7: F01 – Frontend |
| └─ PriceTicker component (homepage horizontal ticker) |
| └─ useSWR client hook with 5-min polling |
| └─ PriceCard, PriceGrid components |
| └─ /energy-prices page (full dashboard) |
| └─ SourceAttribution + PriceMetadata components |
| └─ Responsive styling + loading/empty states |
|
| Day 8-10: F05 – Crawler + Extraction Pipeline |
| └─ Crawler framework (rate limiting, retry, backoff) |
| └─ Tier 1 crawler (6 Algerian official sources) |
| └─ Tier 2-4 crawlers (13 international/news/company sources) |
| └─ AI extraction pipeline (cheerio → gpt-4o-mini → 18 fields) |
| └─ Extraction prompt tuning for FR/AR/EN content |
| └─ Token tracking + daily budget (300K tokens) |
```

WEEK 3 – F05 Deduplication + Database + Review CMS

```
| Day 11-12: F05 – Deduplication + Change Detection |
| └─ Embedding-based name matching (cosine similarity ≥0.85) |
| └─ GPS proximity matching (5 km radius) |
| └─ Multi-signal matching (partners + type + wilaya) |
| └─ Change detection: field-level diff |
| └─ Outcome classification: NEW / UPDATED / UNCHANGED / CONFLICT |
```

- Day 13-14: F05 – Review CMS
 - └ /admin/projects/review page (New | Changes | Conflicts tabs)
 - └ ProjectReviewCard component
 - └ Review actions API (approve, edit, reject, dismiss, resolve)
 - └ Sync history tracking (JSONB audit trail)
 - └ Stale project flagging (>30 days without update)

WEEK 4 – F05 Frontend + F01 Intelligence Pages

- Day 15-16: F05 – Frontend
 - └ /energy-projects page (hero, description, last sync time)
 - └ ProjectFilters component (category pills, status, wilaya)
 - └ ProjectGrid + ProjectCard components
 - └ ProjectDetail (expandable/modal with all 18 fields)
 - └ Source attribution + last-updated display
- Day 17-19: F05 – Interactive Map
 - └ Leaflet + React-Leaflet integration
 - └ GPS markers colored by project status
 - └ Marker clustering for dense areas
 - └ Popup cards (name, status, capacity, link to detail)
 - └ Map/filter sync (filtering updates map markers)
 - └ Responsive map (full width desktop, 300px mobile)
- Day 20-21: F01 – Intelligence Pages
 - └ /intelligence/[benchmark] route
 - └ Price header with full metadata
 - └ Historical chart integration (from energy_prices table)
 - └ Related benchmarks section
 - └ Sources & References section
 - └ Phase 2 shell (AI analysis + news + context = Phase 3)

WEEK 5 – Integration, Testing, Polish

- Day 22-24: Integration + Testing
 - └ Homepage ticker integration
 - └ Cross-linking: intelligence ↔ project details
 - └ Seed 12 initial projects for validation
 - └ End-to-end crawl → extract → review → publish flow
 - └ Mobile responsiveness testing
 - └ Loading states + error boundaries
 - └ i18n strings for new pages (EN/FR/AR)
 - └ Performance check (LCP, CLS, bundle size)
- Day 25: Deployment
 - └ Vercel cron job setup (every 6-12h for F05 crawlers)

- | | Environment variables (EIA key, OpenAI key, extraction cap) |
- | | Staging deploy → test → production deploy |
- | | Documentation handoff |

4.3 Detailed Task Breakdown

Foundation (2 days)

#	TASK	EST.	DEPENDENCIES
F-01	Create <code>energy_prices</code> migration + indexes + materialized view	0.5d	—
F-02	Create <code>energy_projects</code> migration + indexes	0.5d	—
F-03	Install packages (swr, cheerio, recharts, leaflet, react-leaflet, readability)	0.25d	—
F-04	Define TypeScript types (BenchmarkPrice, ProjectSchema, PriceProvider, etc.)	0.5d	F-01, F-02
F-05	Benchmark registry + provider interface scaffolding	0.25d	F-04

Feature 1 — Live Energy Prices (5 days)

#	TASK	EST.	DEPENDENCIES
LP-01	PriceProvider interface + base types	0.25d	F-04
LP-02	OPEC provider implementation (opec.org scraping)	0.5d	LP-01
LP-03	EIA provider implementation (API)	0.5d	LP-01
LP-04	OilPrice provider implementation (fallback scraping)	0.5d	LP-01
LP-05	Argus/Platts provider stub (ready for subscription)	0.25d	LP-01
LP-06	Resolver logic (merge + Saharan Blend → Brent fallback)	0.5d	LP-02-LP-05
LP-07	In-memory cache (5-min TTL, per-tier)	0.25d	LP-06
LP-08	GET <code>/api/energy-prices</code> endpoint (aggregate + cache + DB write)	1d	LP-06, LP-07
LP-09	Historical DB write (energy_prices INSERT on every fetch)	0.25d	LP-08
LP-10	GET <code>/api/intelligence/[benchmark]</code> endpoint (stub for Phase 3)	0.5d	LP-08
LP-11	PriceCard component (price, change%, trend, metadata)	0.5d	LP-06
LP-12	PriceGrid component (responsive grid of cards)	0.25d	LP-11
LP-13	PriceTicker component (horizontal scrolling homepage widget)	0.5d	LP-11
LP-14	<code>/energy-prices</code> page (full dashboard)	0.5d	LP-08, LP-12
LP-15	<code>/intelligence/[benchmark]</code> page (price + chart + sources)	1d	LP-10, LP-11
LP-16	SWR client hook with 5-min polling	0.25d	LP-08
LP-17	Public messaging rules implementation	0.25d	LP-11

#	TASK	EST.	DEPENDENCIES
LP-18	Homepage ticker integration	0.25d	LP-13

Feature 2 — Algeria Project Map Interactive (18-22 days)

Crawler Pipeline (4 days)

#	TASK	EST.	DEPENDENCIES
EP-01	Crawler framework (rate limiter, retry, backoff, User-Agent)	1d	—
EP-02	Tier 1 crawler: 6 Algerian official sources (energy.gov.dz, sonatrach.com, sonelgaz.dz, alnaft.dz, aapi.dz, joradp.dz)	1.5d	EP-01
EP-03	Tier 2 crawler: 5 international org sources (World Bank, IEA, OPEC, GEM, AfDB)	1d	EP-01
EP-04	Tier 3-4 crawler: Reuters + 7 company sources (Eni, TotalEnergies, BP, Equinor, ExxonMobil, Chevron, Oxy)	0.5d	EP-01

AI Extraction Pipeline (3 days)

#	TASK	EST.	DEPENDENCIES
EP-05	Content fetcher (cheerio + @mozilla/readability → clean text)	0.5d	EP-01
EP-06	LLM extraction prompt (18-field JSON, FR/AR/EN, confidence scores)	1d	EP-05
EP-07	Extraction pipeline orchestration (fetch → extract → validate)	1d	EP-05, EP-06
EP-08	Token tracking + daily budget enforcement (300K tokens)	0.5d	EP-07

Deduplication + Change Detection (2 days)

#	TASK	EST.	DEPENDENCIES
EP-09	Embedding generation (text-embedding-3-small) for project names	0.5d	—
EP-10	Name similarity matching (cosine ≥ 0.85)	0.5d	EP-09
EP-11	GPS proximity matching (5 km radius)	0.25d	—
EP-12	Multi-signal matching (partners + type + wilaya overlap)	0.25d	EP-10, EP-11
EP-13	Change detection: field-level diff + outcome classification	0.5d	EP-12

Review CMS (2 days)

#	TASK	EST.	DEPENDENCIES
EP-14	Review queue API endpoints (approve, edit, reject, dismiss, resolve)	1d	EP-13
EP-15	<code>/admin/projects/review</code> page (New / Changes / Conflicts tabs)	1d	EP-14
EP-16	ProjectReviewCard + ReviewQueue components	(in EP-15)	EP-14
EP-17	Sync history tracking (JSONB audit trail)	(in EP-14)	—
EP-18	Stale project flagging (>30 days)	0.25d	EP-13

Frontend (3 days)

#	TASK	EST.	DEPENDENCIES
EP-19	GET /api/energy-projects endpoint (filtered, paginated)	0.5d	F-02
EP-20	ProjectCard component (status badge, category, operators, source)	0.5d	—
EP-21	ProjectFilters component (category pills, status, wilaya, search)	0.5d	—
EP-22	ProjectGrid component (responsive grid of cards)	0.25d	EP-20
EP-23	ProjectDetail component (modal/expand with all 18 fields)	0.5d	EP-20
EP-24	/energy-projects page (hero + filters + grid + map)	0.75d	EP-19-EP-23

Interactive Map (3 days)

#	TASK	EST.	DEPENDENCIES
EP-25	Leaflet + React-Leaflet setup with OpenStreetMap tiles	0.5d	—
EP-26	GPS markers colored by project status	0.5d	EP-25
EP-27	Marker clustering (Leaflet.markercluster)	0.5d	EP-26
EP-28	Popup cards (name, status, capacity, detail link)	0.5d	EP-26
EP-29	Map/filter sync (filter bar updates map markers)	0.5d	EP-26, EP-21
EP-30	Responsive map behavior (mobile collapse, full desktop)	0.5d	EP-25

Integration & Testing (3 days)

#	TASK	EST.	DEPENDENCIES
INT-01	Seed 12 initial projects for validation	0.5d	EP-19
INT-02	End-to-end test: crawl → extract → dedupe → review → publish → display	1d	All
INT-03	Cross-linking: intelligence pages ↔ project details	0.25d	LP-15, EP-23
INT-04	Mobile responsiveness QA	0.25d	All UI
INT-05	i18n strings for new pages (EN/FR/AR)	0.5d	All UI
INT-06	Loading states + error boundaries	0.25d	All UI
INT-07	Performance audit (LCP, CLS, bundle size)	0.25d	All

Deployment (1 day)

#	TASK	EST.	DEPENDENCIES
DEP-01	Vercel cron job: F05 crawlers (every 6-12h)	0.25d	EP-02-EP-04
DEP-02	Environment variables (EIA key, OpenAI key, extraction cap)	0.25d	—
DEP-03	Staging deploy → smoke test → production deploy	0.5d	All

4.4 Milestones

MILESTONE	WHEN	DELIVERABLE
M1 — Foundation Ready	Day 2	DB tables, types, registry, packages installed
M2 — Prices API Live	Day 5	<code>/api/energy-prices</code> returns 7 benchmarks with metadata
M3 — Price UI Complete	Day 7	Ticker on homepage, <code>/energy-prices</code> dashboard, intelligence pages
M4 — Crawlers Running	Day 10	All 19 sources crawled, data flowing to extraction pipeline
M5 — Review CMS Working	Day 14	Admin can approve, edit, reject, resolve conflicts
M6 — Map Interactive	Day 19	Leaflet map with GPS markers, clustering, filter sync
M7 — Integration Done	Day 22	Cross-linking, responsiveness, i18n, loading states
M8 — Deployed to Prod	Day 24-25	Staging tested → production live, cron running

4.5 Risk Register

RISK	IMPACT	LIKELIHOOD	MITIGATION
Argus/Platts subscription not acquired	Medium	Medium	All providers built as pluggable; deploy with OPEC+EIA+OilPrice fallback. Swap in Argus/Platts when subscribed. No code changes needed.
Algerian gov sites slow/unavailable	Medium	High	Conservative crawl rate with exponential backoff; cache last known data; flag in review queue
AI extraction hallucinates project data	High	Medium	Confidence scoring per field; human review queue; cross-source verification; never auto-publish
Leaflet map performance with 50+ markers	Low	Low	Marker clustering; only render visible markers
Crawl costs exceed budget (OpenAI tokens)	Medium	Low	Daily token cap (300K); conservative crawl frequency (6-12h); gpt-4o-mini is cheap
Vercel cron 300s timeout on full crawl	Medium	Medium	Split crawl across multiple cron jobs (Tier 1 separate from Tier 2-4); use Pro plan if needed
Chart library bundle size bloats page	Low	Low	Use recharts (tree-shakeable); lazy load intelligence page charts
Stale data displayed as current	Medium	Medium	"Last updated" timestamp always visible; stale flagging at 30 days; "Temporarily unavailable" for failed fetches

4.6 Key Architecture Decisions

DECISION	RATIONALE
Provider-agnostic API facade	Swap Argus ↔ Platts ↔ any future source without frontend changes
In-memory cache before API response	Avoids DB round-trip for every client poll (every 5 min)
DB persist on every fetch anyway	Builds historical dataset for future analytics + AI
Crawl every 6-12h, not real-time	Gov sites are slow; project data changes weekly/monthly
AI extraction via LLM, not deterministic scraping	Pages are unstructured, mixed-language (FR/AR/EN), diverse formats
Review queue, not auto-publish	Credibility demands human oversight — CMS is for review, not data entry
Intelligence pages as Phase 2 shell	Price + chart + sources now; AI analysis + news = Phase 3 when RAG mature
Public messaging: "Temporarily unavailable"	Never expose internal licensing/infrastructure details
19 sources locked for v1	Scope control; new sources = future phase

Appendices

A. New Database Migrations

Migration 1: `energy_prices` table

```
-- supabase/migrations/xxxxx_create_energy_prices.sql
CREATE TABLE energy_prices (
  id          BIGSERIAL PRIMARY KEY,
  benchmark   VARCHAR(64)  NOT NULL,
  category    VARCHAR(32)  NOT NULL,
  value       DECIMAL(12,4) NOT NULL,
  unit        VARCHAR(16)  NOT NULL,
  change      DECIMAL(8,4),
  change_pct  DECIMAL(6,2),
  trend       VARCHAR(4),
  provider    VARCHAR(32)  NOT NULL,
  provider_tier SMALLINT    NOT NULL,
  delay_status VARCHAR(20) NOT NULL,
  confidence  VARCHAR(8)   NOT NULL,
  availability VARCHAR(24) NOT NULL DEFAULT 'available',
  source_url   TEXT,
  fetched_at   TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  is_active_primary BOOLEAN DEFAULT FALSE
);

CREATE INDEX idx_ep_benchmark_time ON energy_prices (benchmark, fetched_at DESC);
CREATE INDEX idx_ep_category_time ON energy_prices (category, fetched_at DESC);
CREATE INDEX idx_ep_fetched_at ON energy_prices (fetched_at DESC);

CREATE MATERIALIZED VIEW energy_prices_latest AS
SELECT DISTINCT ON (benchmark)
  benchmark, value, unit, change, change_pct, trend,
  provider, provider_tier, delay_status, confidence, fetched_at
FROM energy_prices
WHERE fetched_at > NOW() - INTERVAL '2 hours'
ORDER BY benchmark, fetched_at DESC;
```

Migration 2: `energy_projects` table

```
-- supabase/migrations/xxxxx_create_energy_projects.sql
CREATE TABLE energy_projects (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  name TEXT NOT NULL,
  slug TEXT NOT NULL UNIQUE,
  energy_sector TEXT NOT NULL CHECK (energy_sector IN
('oil', 'gas', 'lng', 'renewable', 'hydrogen', 'power', 'petrochemical', 'other')),
  project_type TEXT CHECK (project_type IN
('pipeline', 'field', 'plant', 'terminal', 'grid', 'solar', 'wind', 'exploration', 'other'))
,
  wilaya TEXT,
  municipality TEXT,
  gps_lat DECIMAL(10,7),
  gps_lng DECIMAL(10,7),
  operating_company TEXT,
  project_partners TEXT[] DEFAULT '{}',
  investment_value TEXT,
  production_capacity TEXT,
  project_status TEXT NOT NULL CHECK (project_status IN
('operational', 'under_construction', 'planned', 'proposed', 'completed', 'suspended')),
  completion_percentage DECIMAL(5,2),
  start_date DATE,
  expected_completion_date DATE,
  project_description TEXT,
  source_url TEXT,
  source_name TEXT,
  last_updated TIMESTAMPTZ DEFAULT NOW(),
  last_source_check TIMESTAMPTZ DEFAULT NOW(),
  source_urls TEXT[] DEFAULT '{}',
  review_status TEXT NOT NULL DEFAULT 'pending' CHECK (review_status IN
('pending', 'approved', 'rejected', 'changes_pending')),
  reviewed_by TEXT,
  reviewed_at TIMESTAMPTZ,
  featured BOOLEAN DEFAULT false,
  sort_order INTEGER DEFAULT 0,
  sync_history JSONB DEFAULT '[]',
  created_at TIMESTAMPTZ DEFAULT NOW(),
  updated_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE INDEX idx_energy_projects_sector ON energy_projects(energy_sector);
CREATE INDEX idx_energy_projects_status ON energy_projects(project_status);
CREATE INDEX idx_energy_projects_wilaya ON energy_projects(wilaya);
CREATE INDEX idx_energy_projects_review ON energy_projects(review_status);
CREATE INDEX idx_energy_projects_source ON energy_projects(source_name);
```

B. New Environment Variables

```
# F01 – Live Prices
# EIA_API_KEY=***          (future: EIA API v2)

# F05 – Project Extraction
EXTRACTION_DAILY_MAX_TOKENS=300000
OPENAI_EMBEDDING_MODEL=text-embedding-3-small
```

C. New Packages

```
{
  "swr": "^2.2.0",
  "cheerio": "^1.0.0",
  "recharts": "^2.12.0",
  "leaflet": "^1.9.0",
  "react-leaflet": "^4.2.0",
  "@types/leaflet": "^1.9.0",
  "@mozilla/readability": "^0.5.0",
  "gpt-tokenizer": "^2.1.0"
}
```

D. PRD References

- [PRD-01: Live Energy Price — Full Specification](#)
- [PRD-05: Algeria Energy Projects Database — Full Specification](#)
- [PRD Index — All Phase 2 Features](#)

Document prepared by: OpenClaw AI Assistant **Next step:** Review with developer (Junaedi) + client (Mr. Ramdane) before starting development.